

Chronic Inflammation

Causes

- **1) PERSISTENCE of Infection**
- **2) PROLONGED EXPOSURE to insult**

Chronic inflammation

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graph TD; A[Chronic inflammation] --> B[Non specific]; A --> C[Granuloma]
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Non specific

Granuloma

Inflammatory cells

Acute

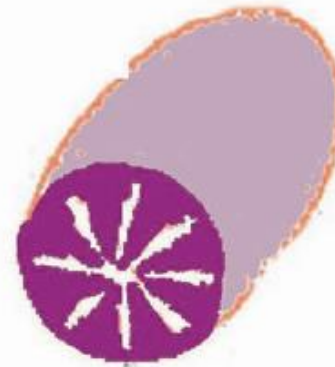


Chronic

Neutrophils

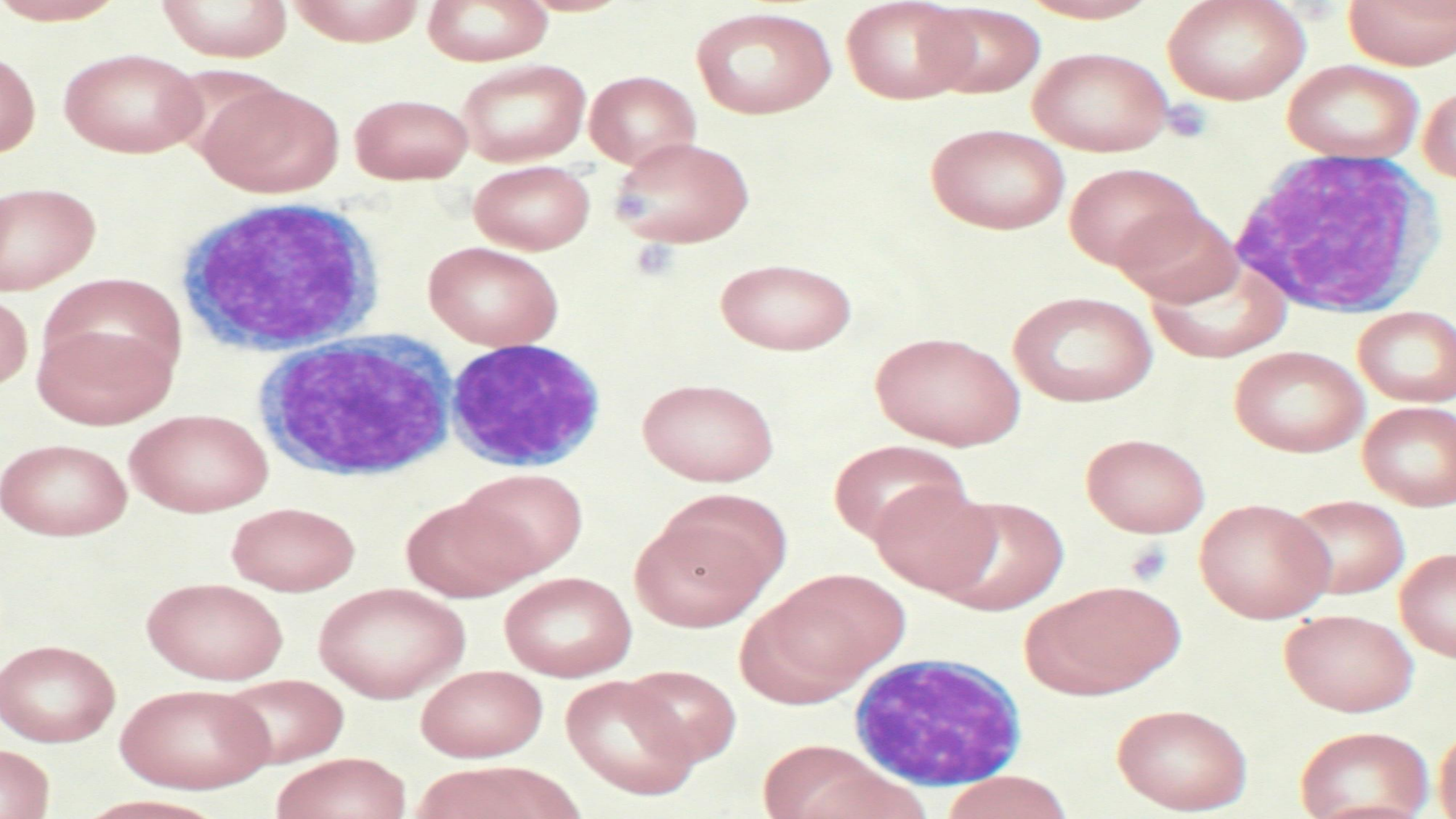


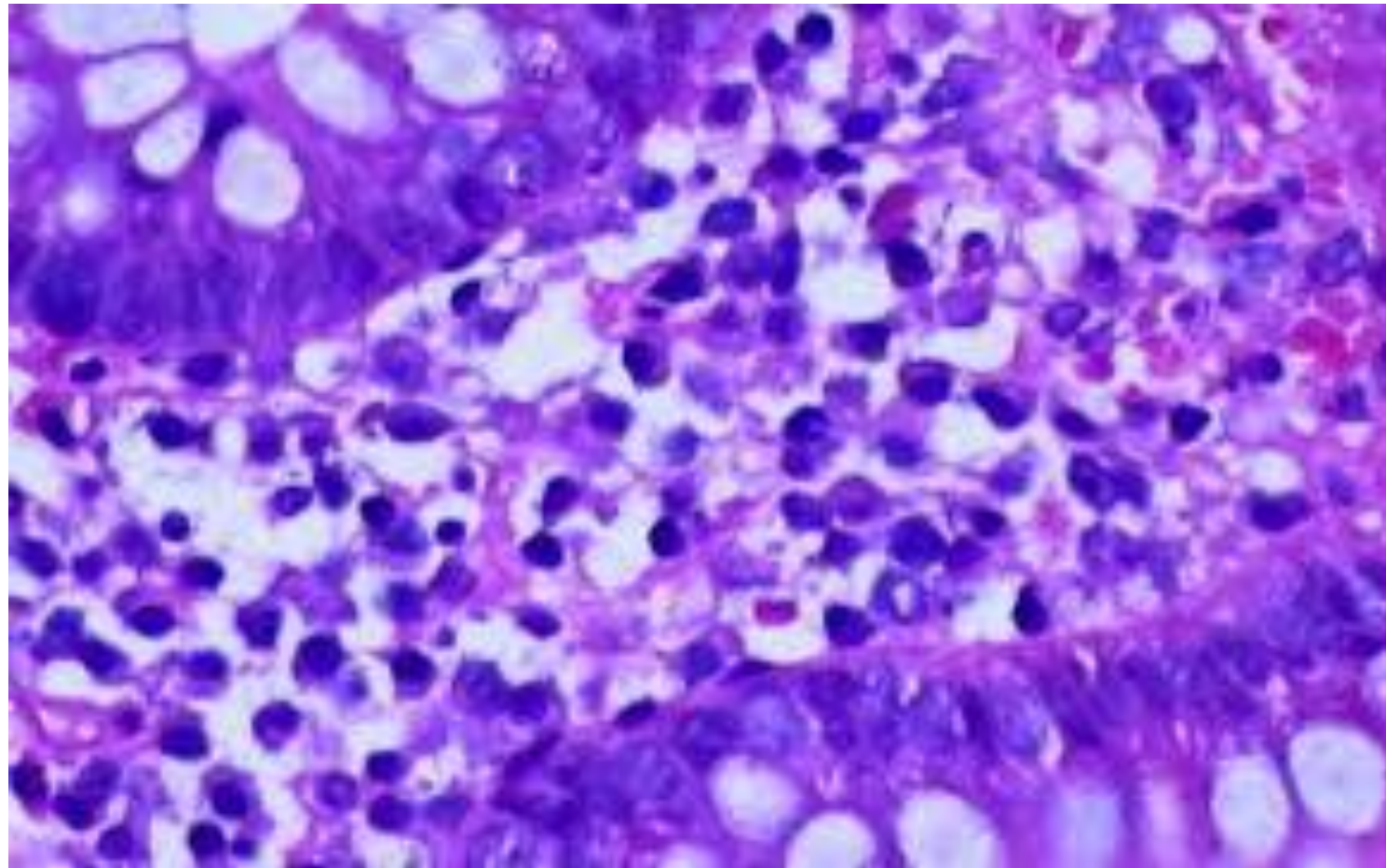
Lymphocyte

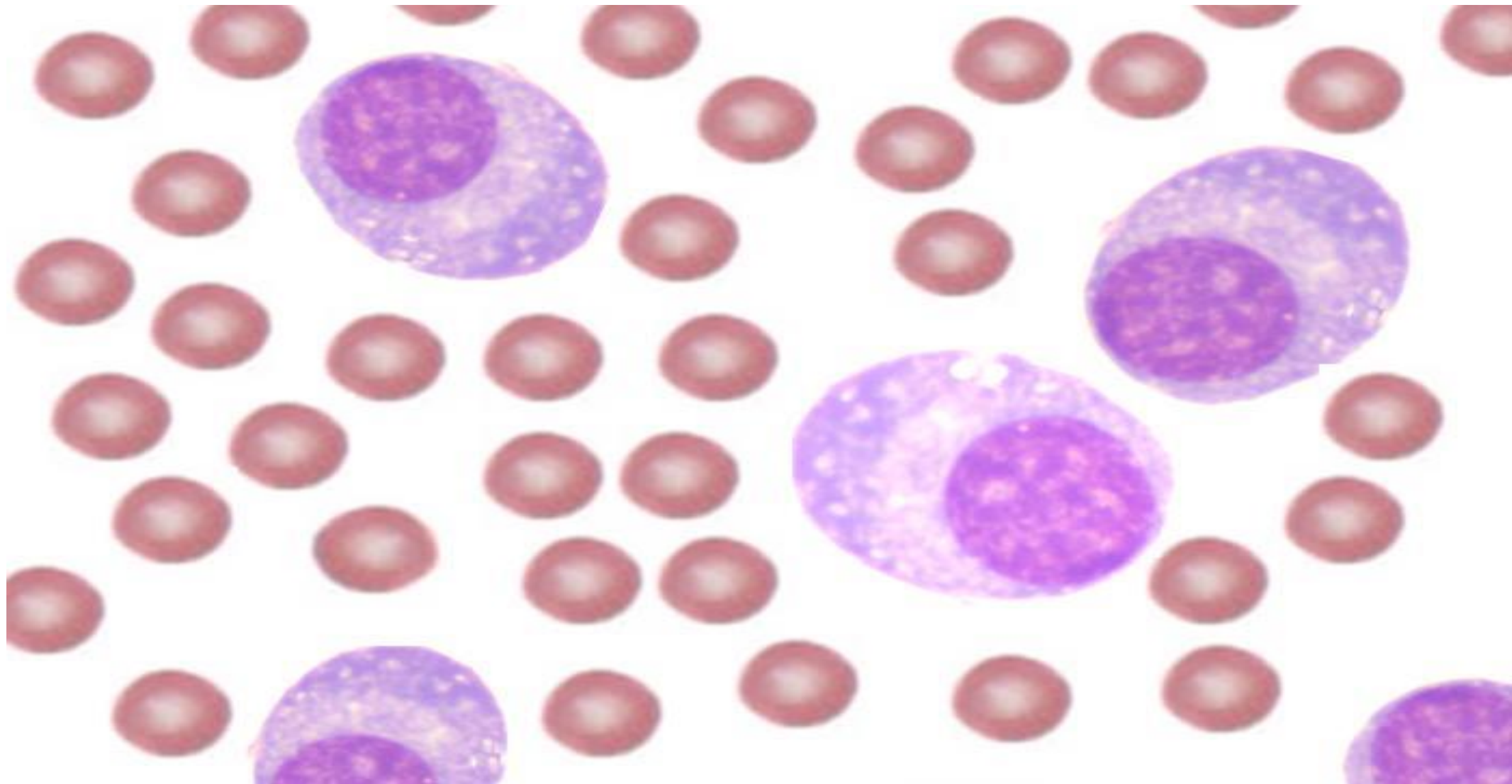


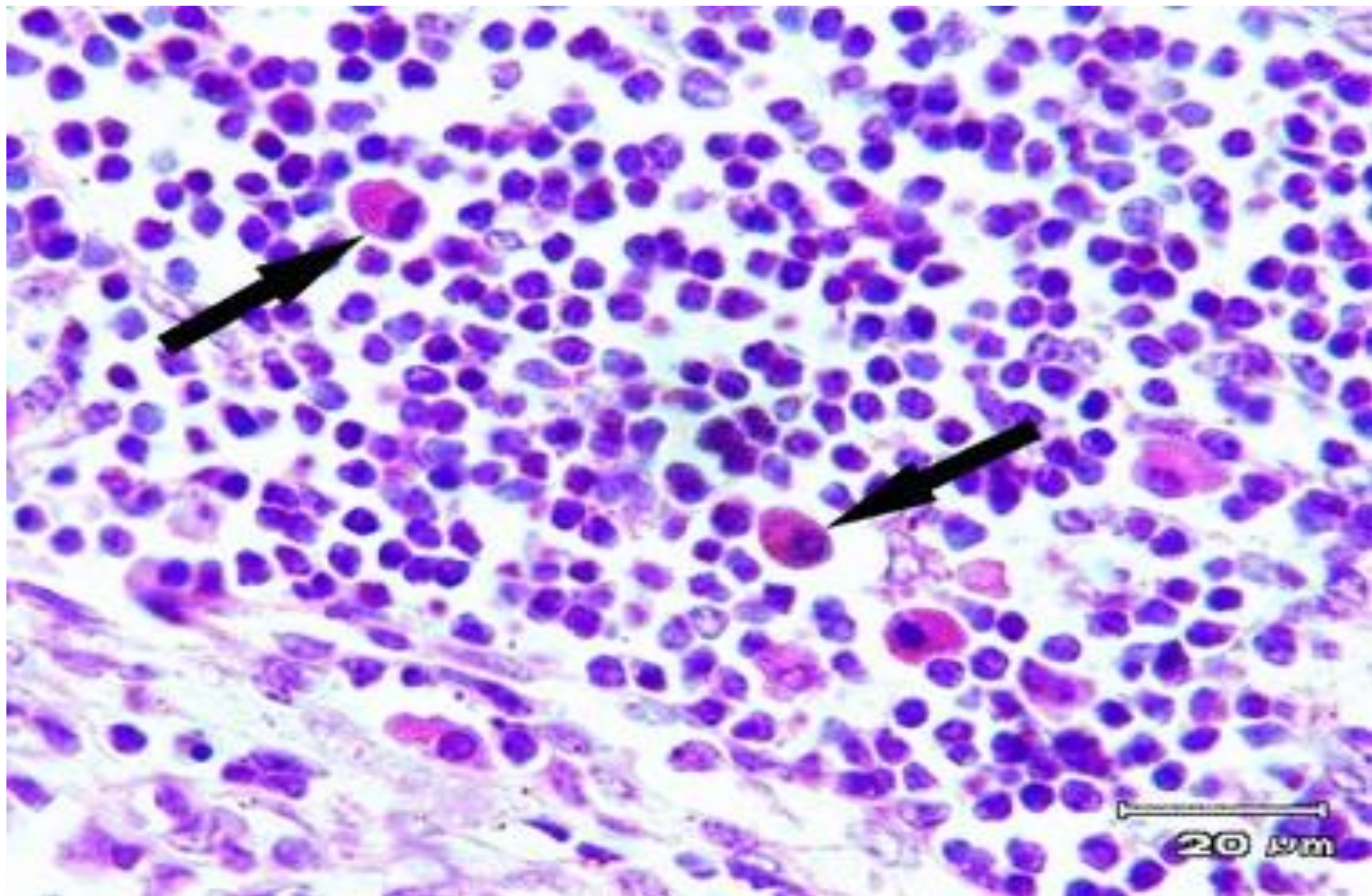
Plasma cell

- Chronic inflammatory cells

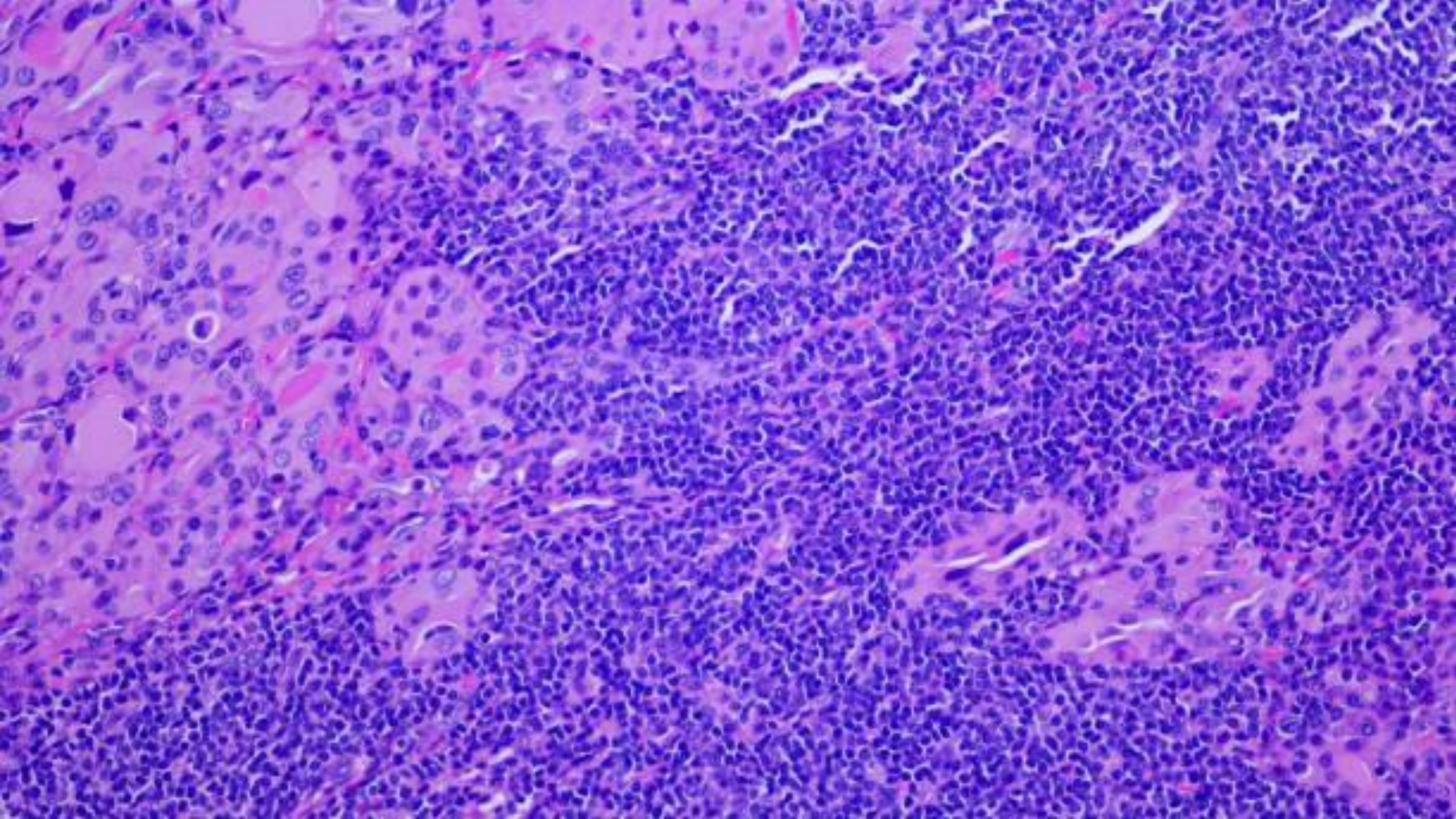


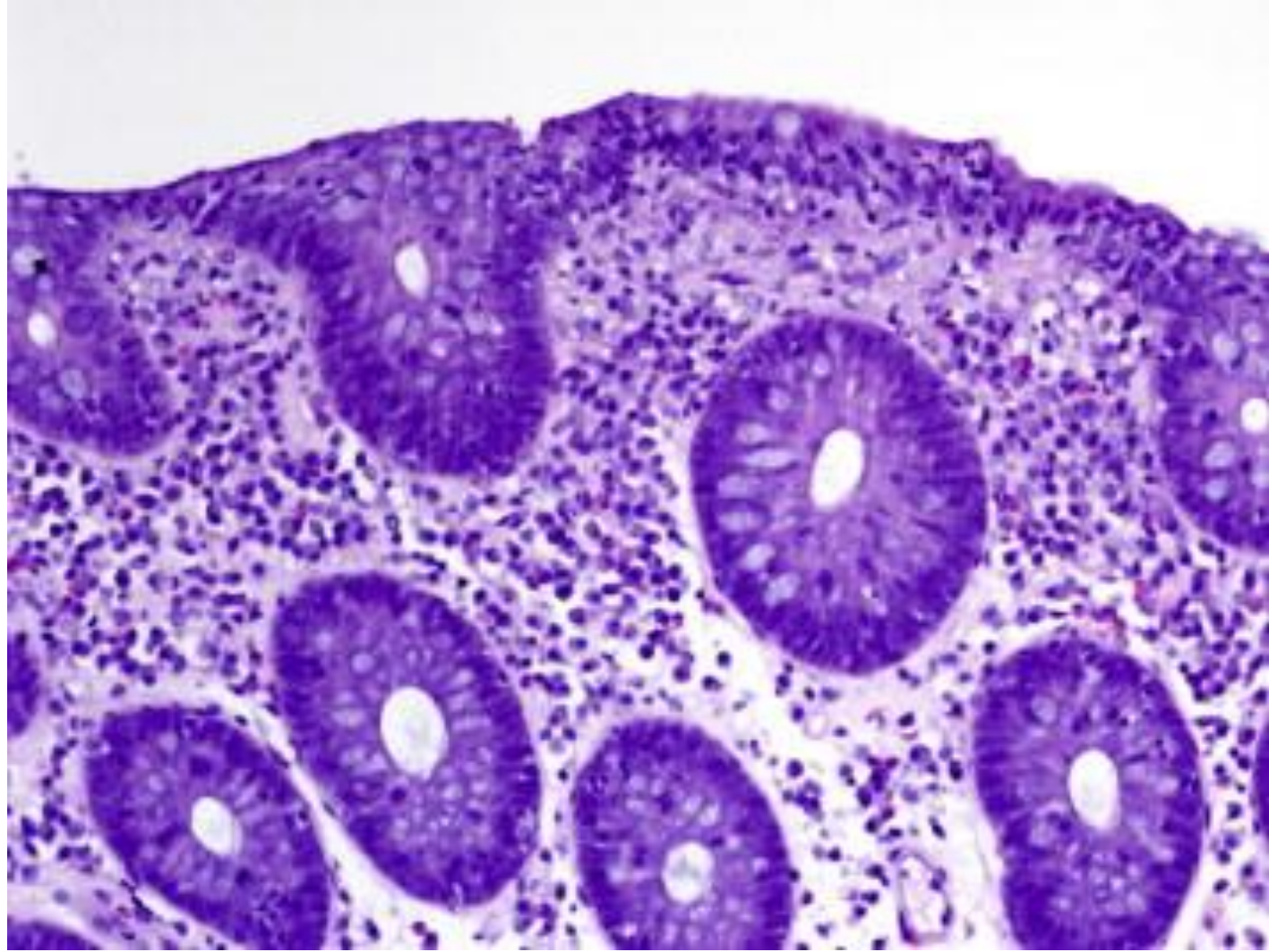






- Chronic non-specific inflammation

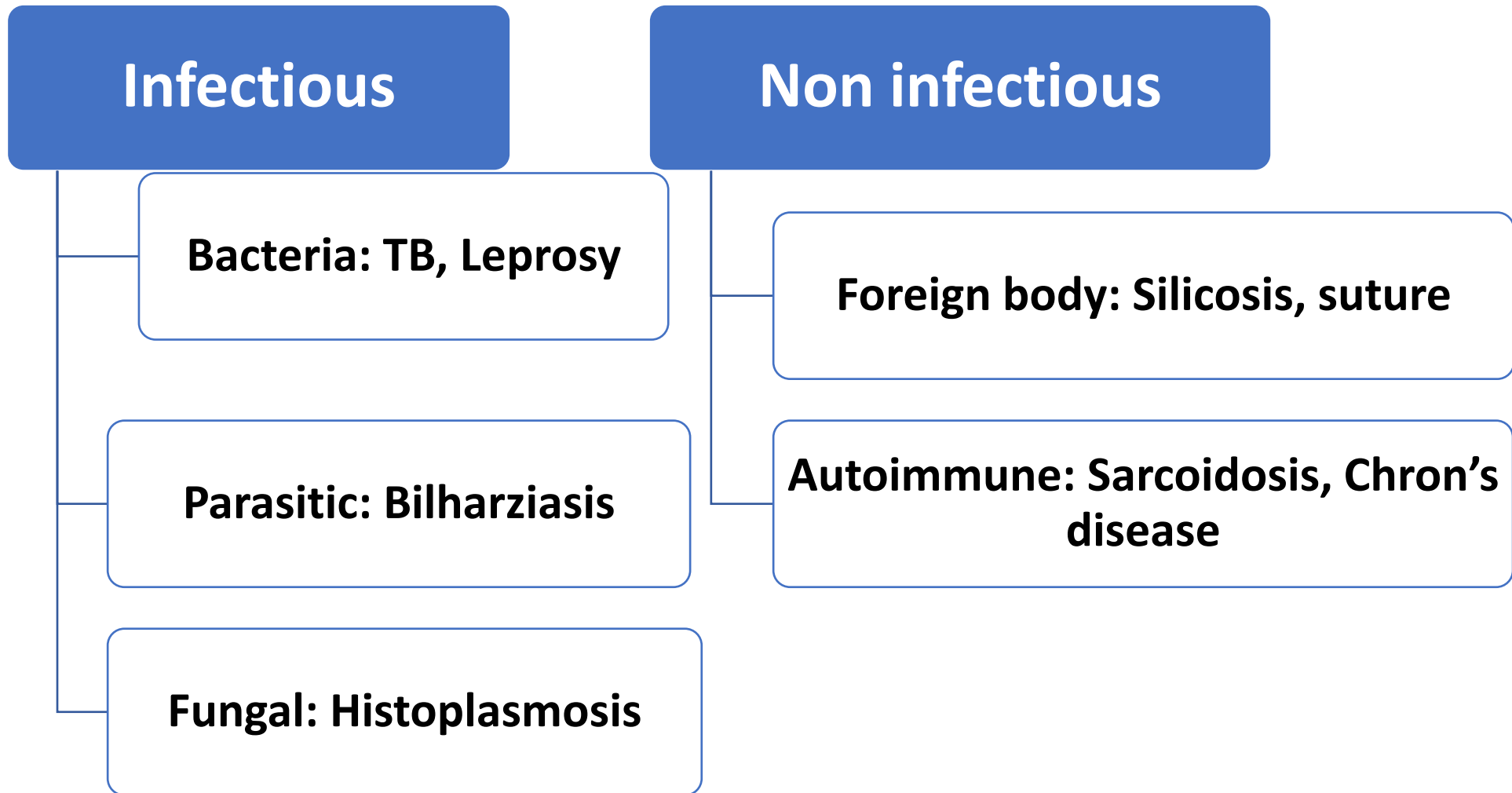




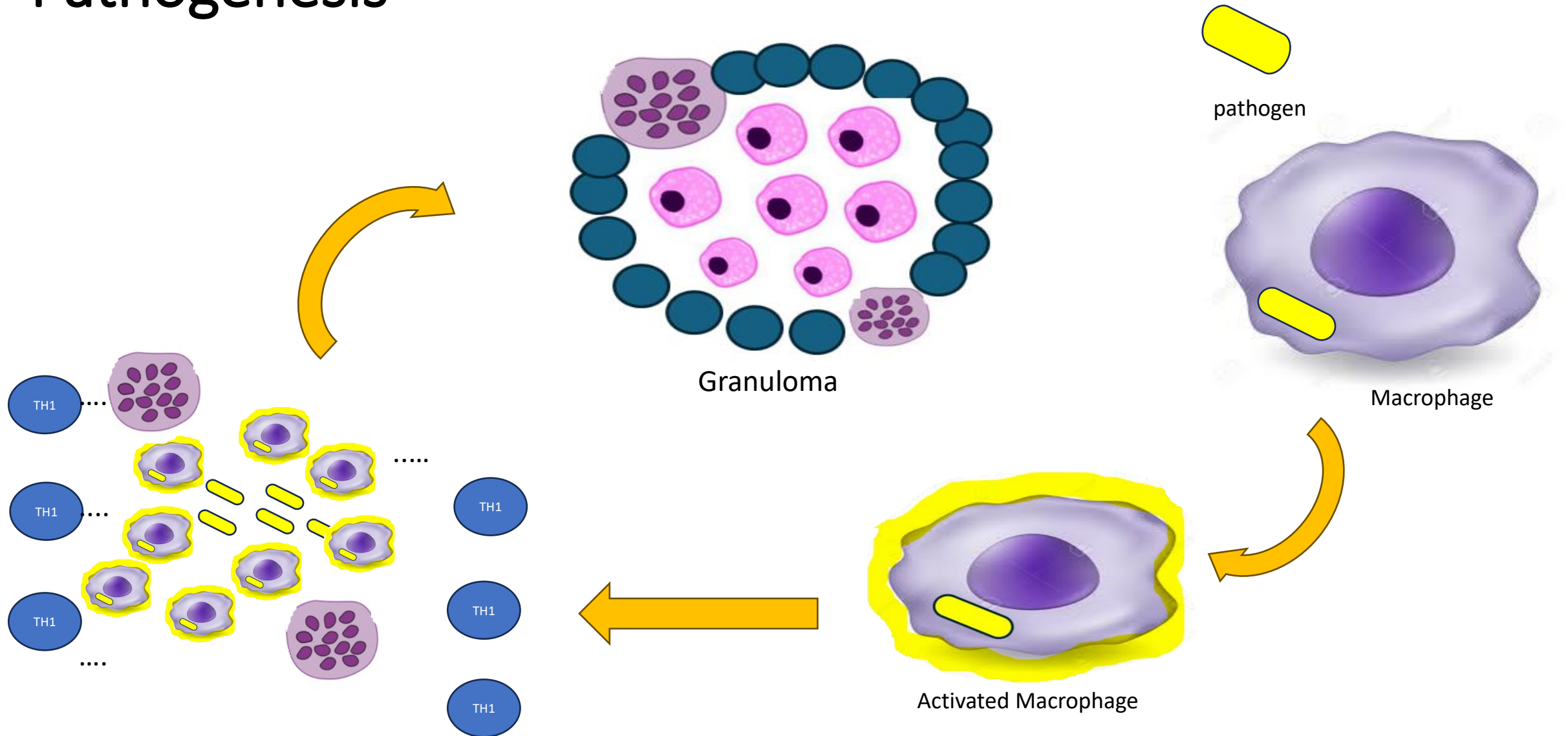
Granuloma

- It is a chronic specific inflammation characterized by focal accumulation of large number of modified macrophages (epitheloid cells) together with lymphocytes, plasma cells, giant cells and fibroblasts forming tumor like mass.

Aetiology

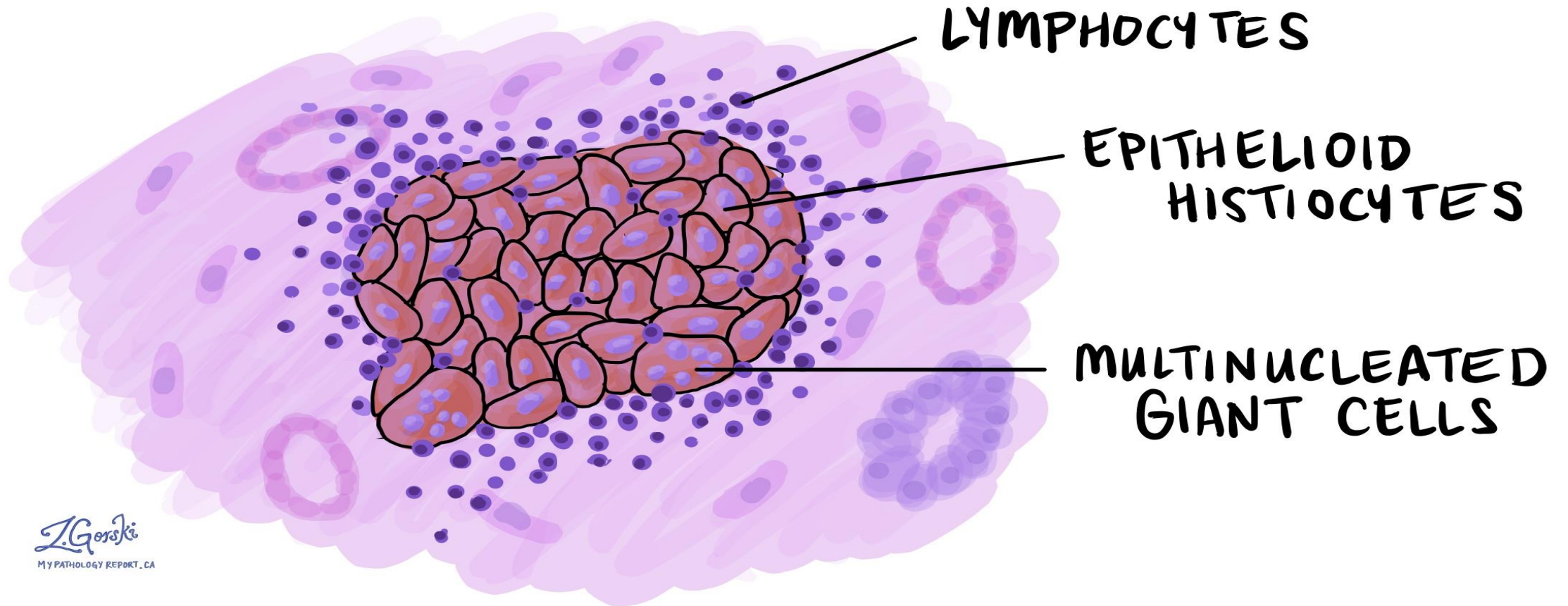


Pathogenesis

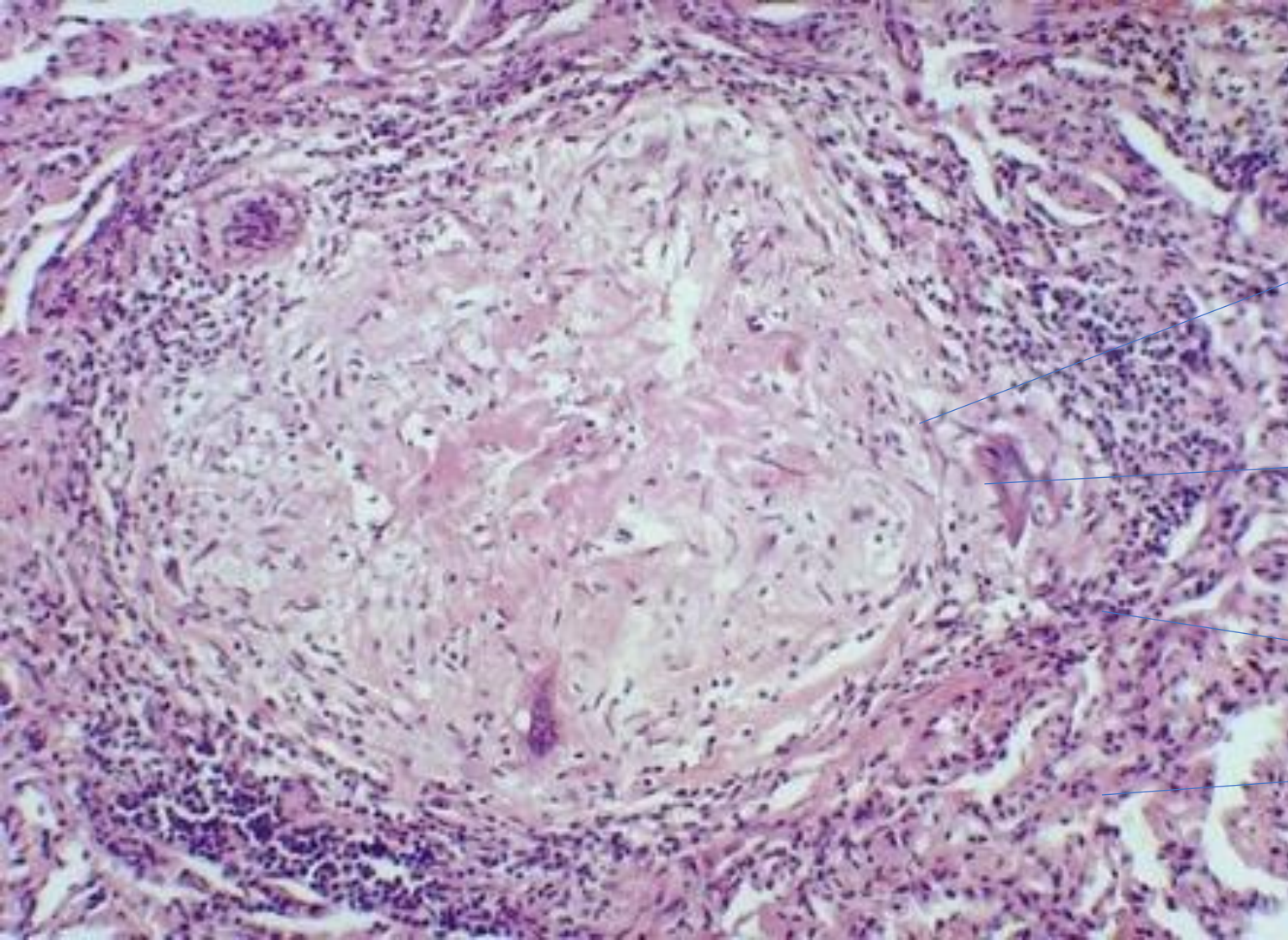


Histological features

GRANULOMA

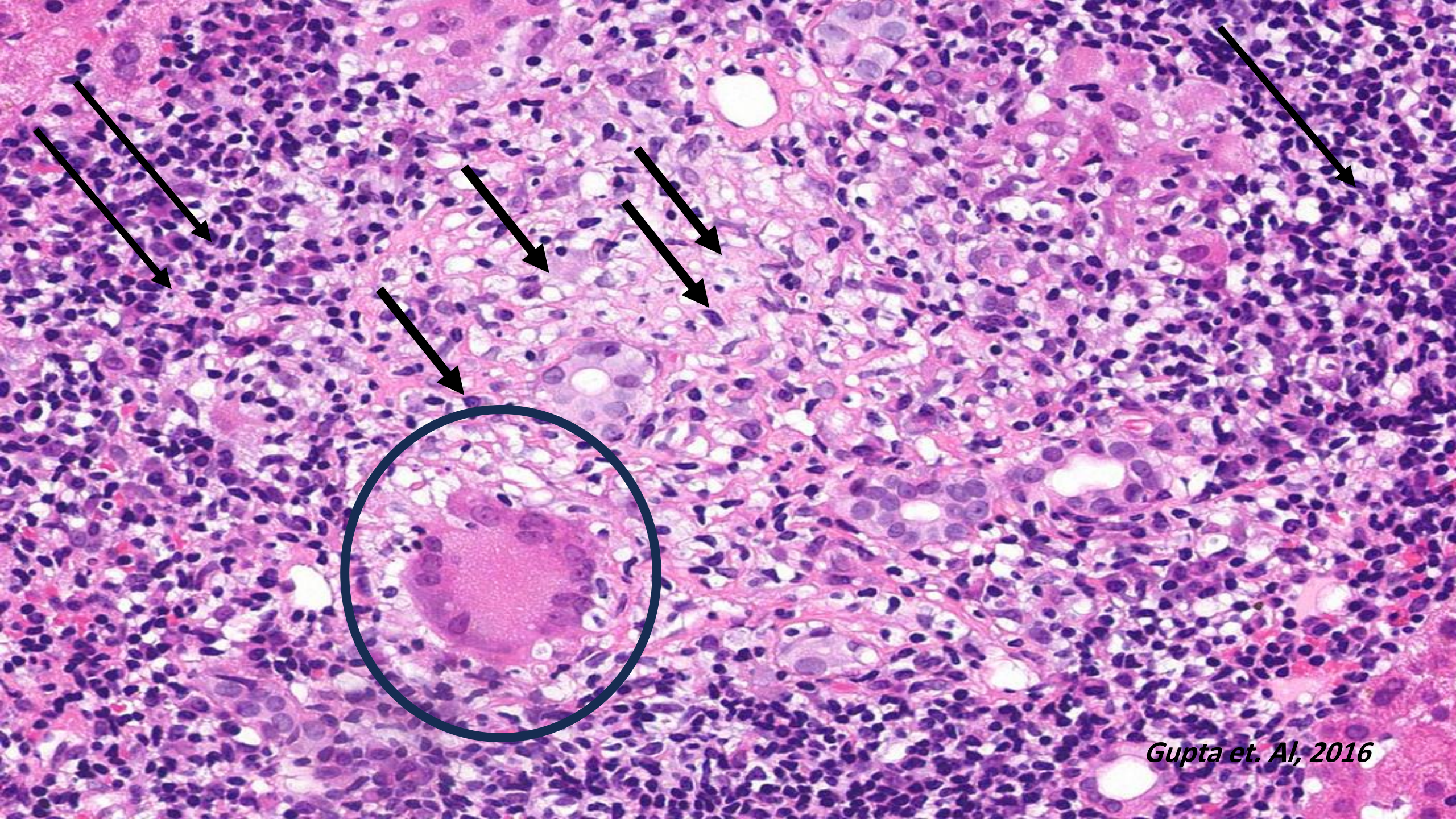


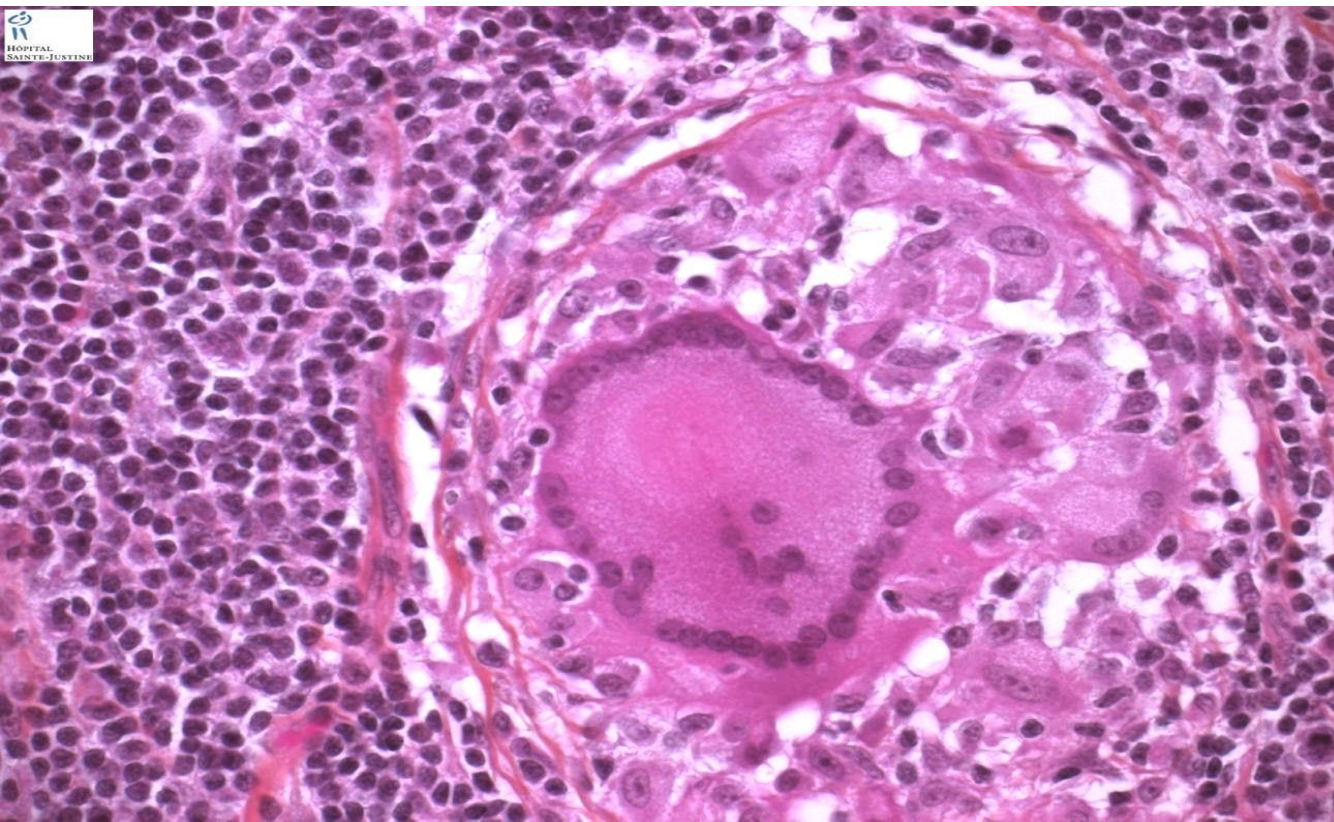
Components of Granuloma



- 4 COMPONENTS

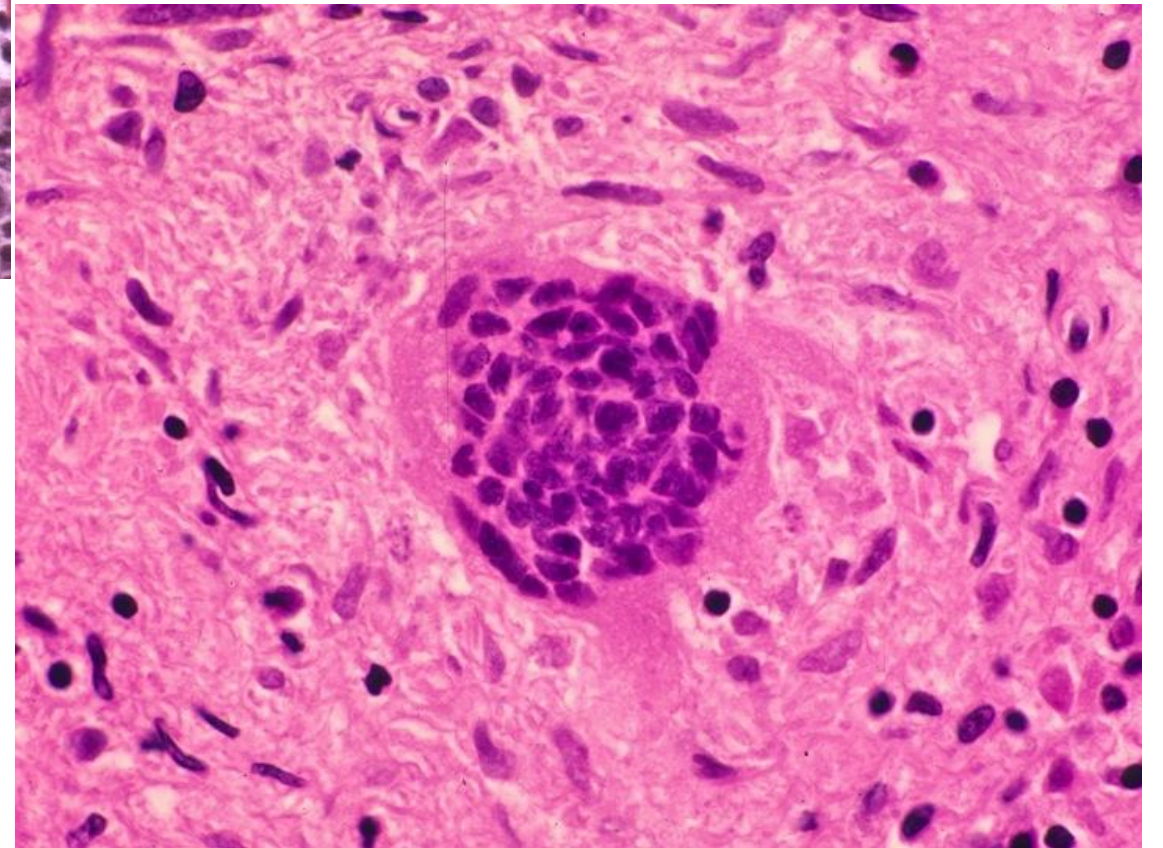
- HISTIOS, epithelioid cells (main component)
- “GIANT” CELLS
Langhans’ FB
- LYMPHS
- FIBROBLASTS





Langhans giant cell

Foreign body giant cell



Tuberculosis

- **Definition:** Chronic infective granulomatous inflammation caused by **tubercle bacilli**.
- **Predisposing factors:** Low standard of nutrition and housing → low natural immunity.

Aetiology

- Mycobacterium tuberculosis.
- Is **acid-fast bacilli** best stained with **Ziehl Nelson stain**.
- Formed of **central core of protein** (tuberculo-protein), **polysaccharide fraction** and **lipid capsule**.

Mode of transmission

-**The human type:** transmitted from human to human by inhalation, ingestion, or inoculation.

-**The bovine type:** transmitted from cows to human by ingestion of infected milk.

Pathogenesis

First 3 weeks → tubercle bacilli enter macrophages by endocytosis → replicate

(Most people → A symptomatic).

Pathogenesis

After 3 weeks → activate macrophages to kill bacteria
(epithelioid macrophages) → release cytokines from
lymphocytes (type VI hypersensitivity) → caseous
necrosis (Cell death) in the center.

Morphology

- Grossly →

Scattered tan-yellow
granulomas



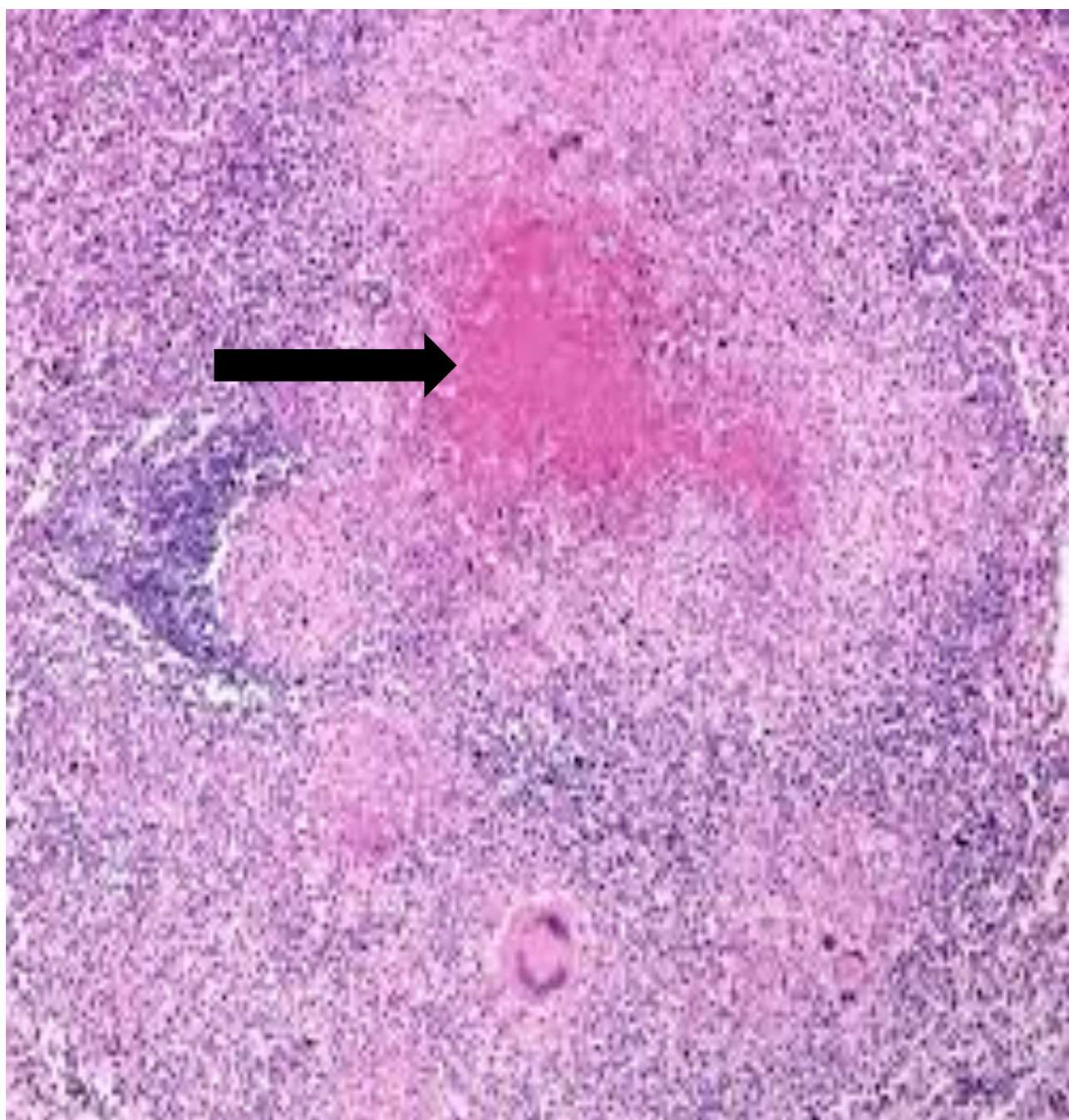
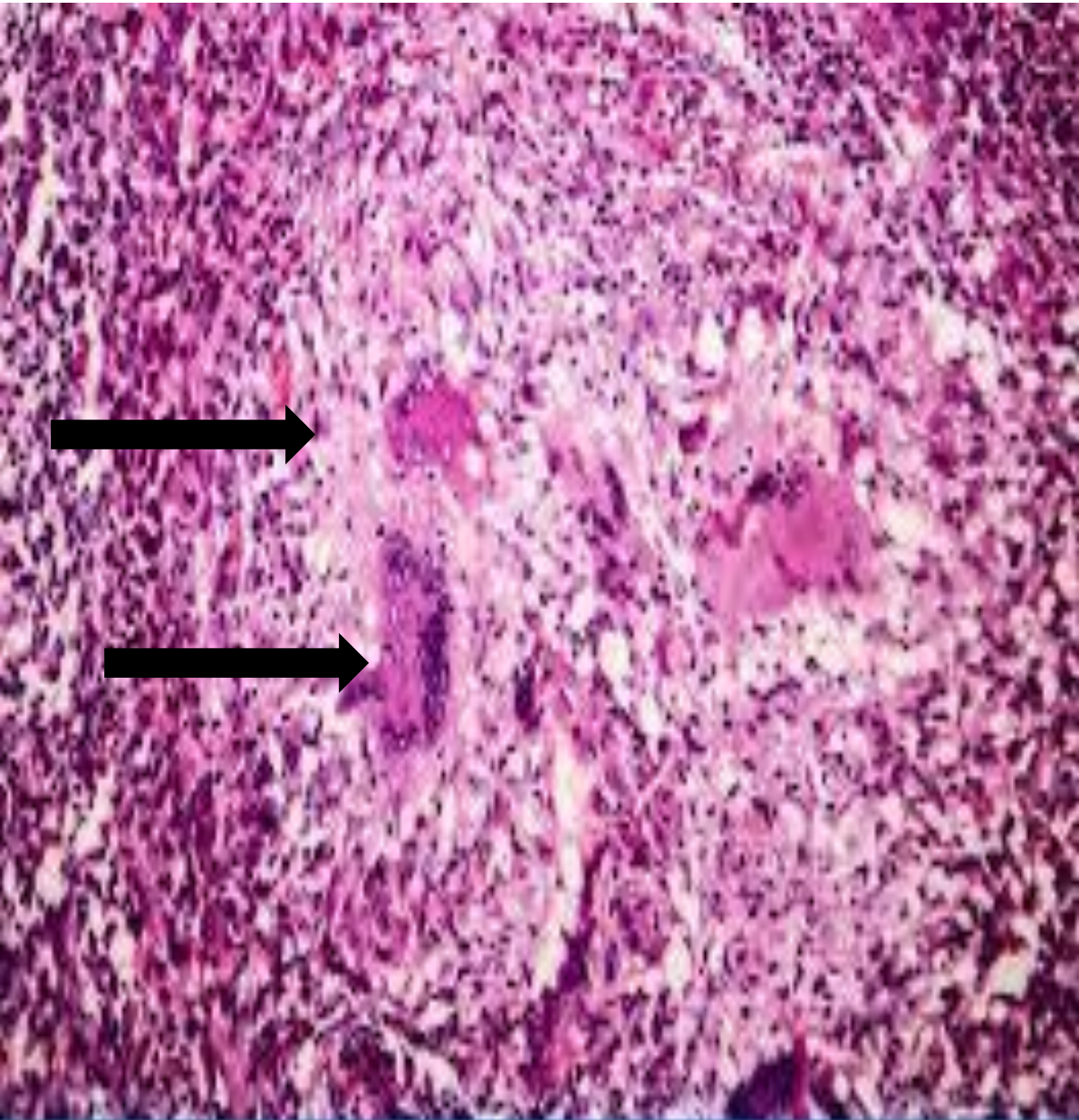
Morphology

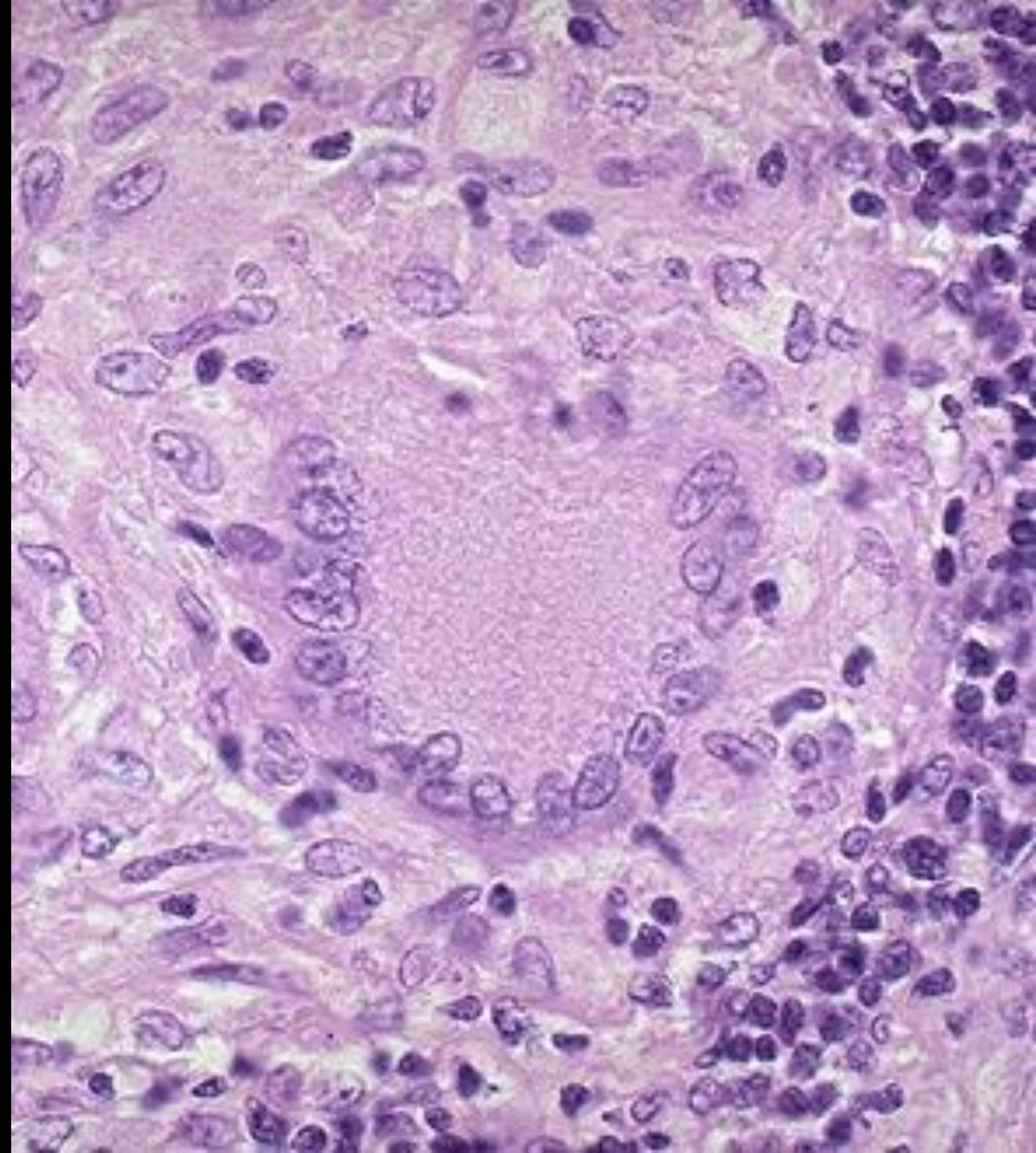
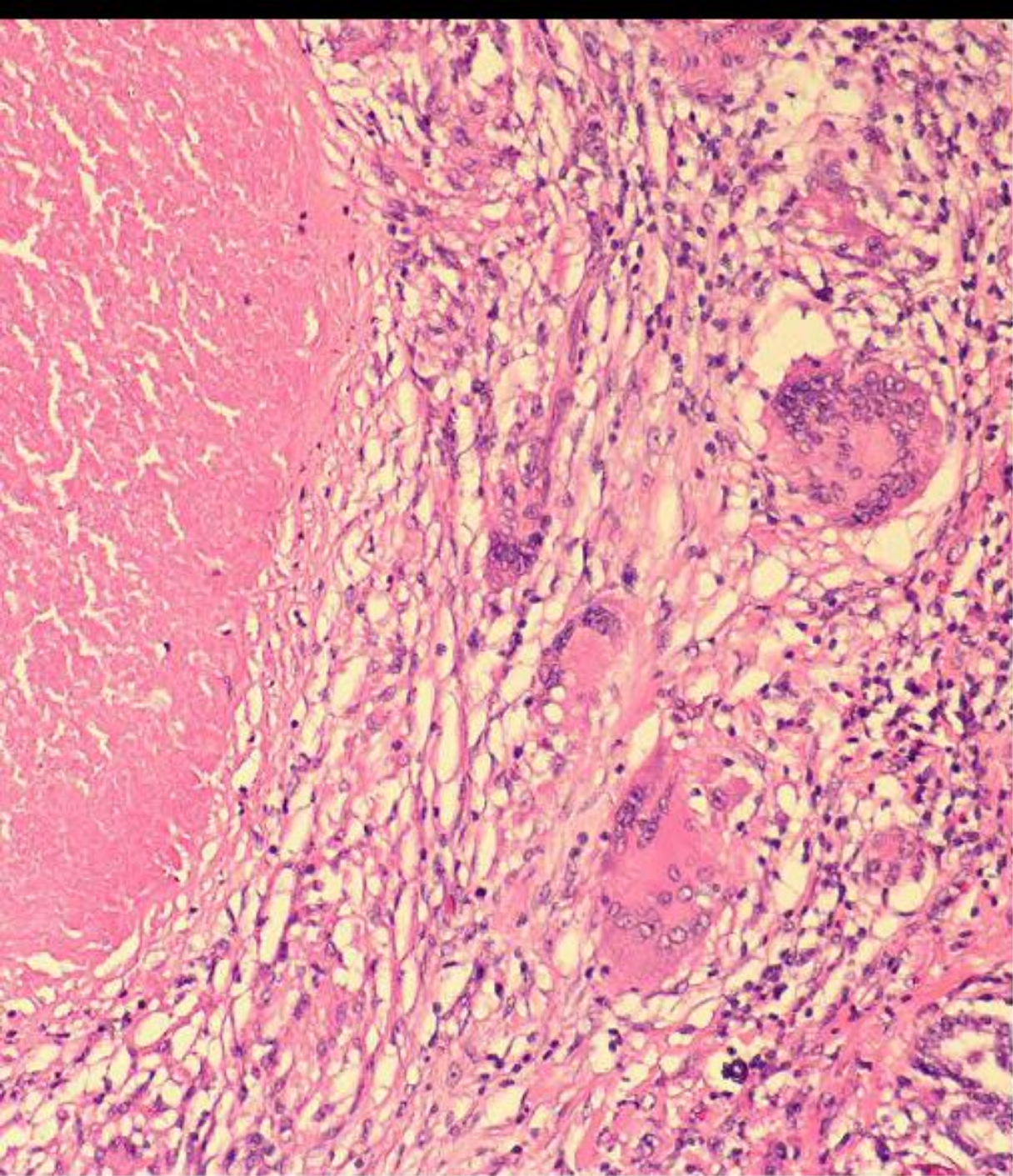
- **Microscopically**

Granuloma

Collection of epithelioid cells (pale cytoplasm and vesicular nuclei), Langhan's multinucleated giant cells (fused epithelioid cells with nuclei arranged in circle or half circle pattern) and lymphocytes with fibrosis surrounding it.

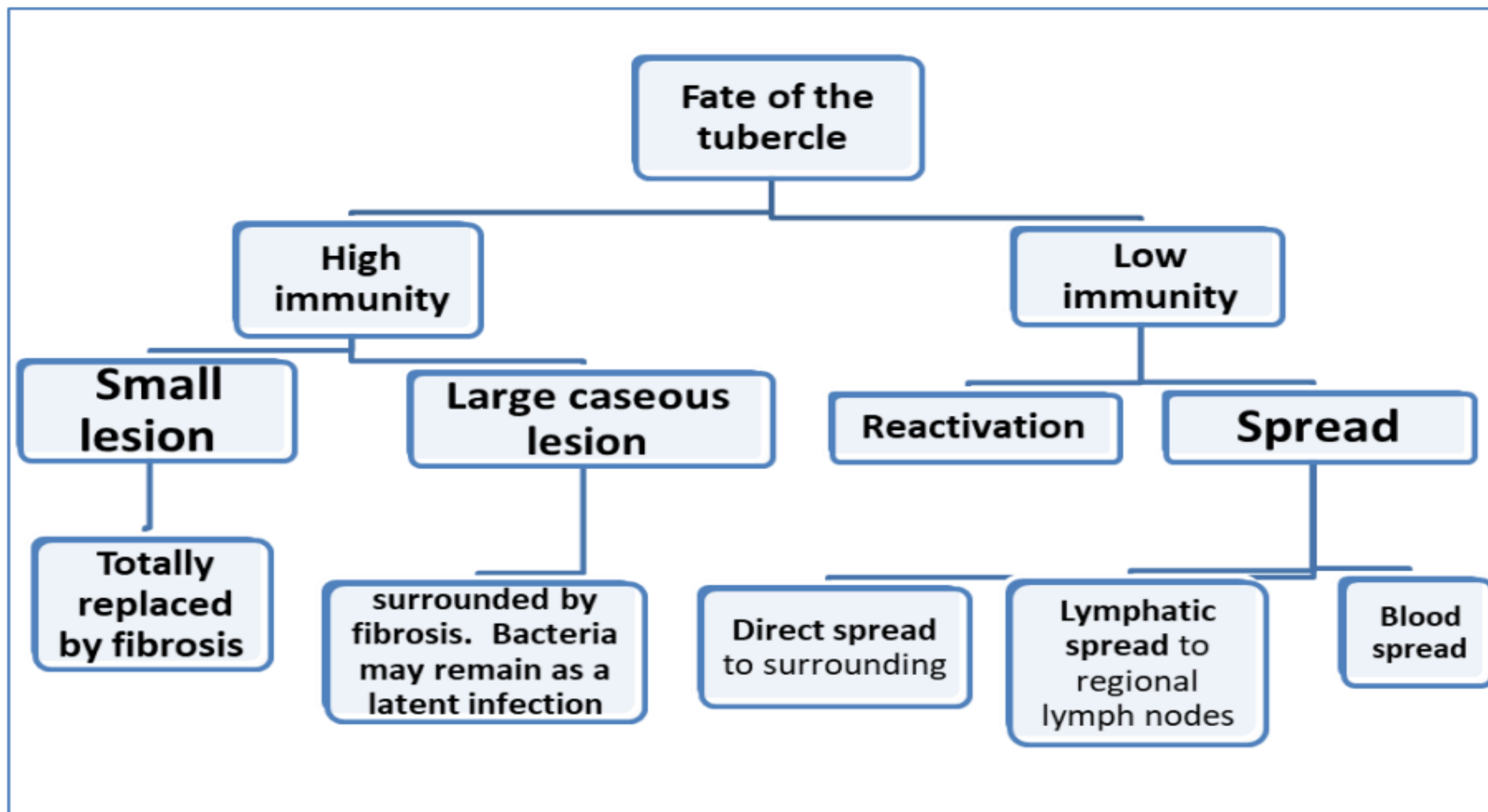
☐ The center of granuloma → caseation necrosis (cheesy grayish yellow material) → structureless homogenous eosinophilic material.





Primary tuberculosis

- Develops in **unexposed (unsensitized) persons** usually children or rarely immuno-suppressed persons who lose their immunity to tubercle bacilli.
- Usually affects primary sites
- ☐ **Inhalation** of infected material leads to **pulmonary tuberculosis**
- ☐ **Ingestion** of infected food leads to **tonsil or intestinal tuberculosis**
- ☐ **Inoculation** of wound leads to **skin lesions**.



Secondary tuberculosis

Arises in **previously sensitized person** (usually adult) either by re-infection or reactivation of dormant bacteria.

May occur with development of secondary pulmonary tuberculosis years after due to:

- lowered immunity (e.g. AIDS)
- Malignant tumors e.g. Hodgkin lymphoma
- Immunosuppressant drugs

	Primary TB	Secondary TB
Age of Exposure	Infection for the firsttime Usually, children	Re-infection or re-activation Usually adult
Organs affected	Primary sites pulmonary,intestine, tonsil, or skin	Any organ
Lymphadenitis	Prominent	Minimal
Tissue destruction	Less marked	More marked (hypersensitivity)
Tissue reaction	Slow	Accelerated

Clinical picture

- History of contact with TB patient
- Toxic manifestation: malaise, anorexia, weight loss, low grade remittent fever, night sweating
- Productive cough, pleuritic pain, haemoptysis



Diagnosis

- **Tuberculin test** (intradermal injection of purified protein derivative of tubercle bacilli that induce localized indurations indicating presence of hypersensitivity reaction)
- **Biopsy:** Characteristic but not pathognomonic granuloma
- **Identification of TB bacilli** (sputum, culture, PCR) **(Most accurate for diagnosis).**

THANK
YOU!